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Authors:
G. Mirsky E. Ruffini H. Nydell R. Foote W. Hawkins
Ciena Corporation OutSys Cisco Systems Nokia University of Cincinnati

RFC 10052

Performance Measurement with Asymmetrical Traffic Using the Simple Two-Way Active Measurement Protocol (STAMP)

Abstract

This document defines an optional extension to the Simple Two-way Active Measurement Protocol (STAMP) that enables a Session-Reflector to send asymmetrical packets, that is, response packets whose size or quantity differs from those sent by the Session-Sender. While standard STAMP exchanges are symmetrical, certain measurement scenarios benefit from reflected packets of different lengths or additional responses to better approximate application traffic conditions. The extension specifies the Reflected Test Packet Control TLV and associated procedures, analyzes challenges in active performance measurement (including in multicast environments), and describes STAMP behaviors to improve measurement efficiency and reduce network impact.

Status of This Memo

This is an Internet Standards Track document.

This document is a product of the Internet Engineering Task Force (IETF). It represents the consensus of the IETF community. It has received public review and has been approved for publication by the Internet Engineering Steering Group (IESG). Further information on Internet Standards is available in Section 2 of RFC 7841.

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1. Introduction

The Simple Two-way Active Measurement Protocol (STAMP) [RFC8762] defines the base STAMP functionalities. STAMP Optional Extensions [RFC8972] introduces a TLV structure that allows a Session-Sender to include optional instructions for Session-Reflectors to extend the functionality of the base STAMP protocol. New STAMP TLVs can be defined to support scenarios like the ones described in [RFC7497], which discusses the coordination of messaging between the source and destination to help deliver one of the fundamental principles of IP performance metric measurements, minimizing the test traffic effect on user flows.

By default, a STAMP Session-Sender and a Session-Reflector exchange packets symmetrically: The number of packets sent by the Session-Reflector and the Session-Sender are the same, and the length of the packets sent by the Session-Reflector and the Session-Sender are the same. However, in some scenarios, e.g., rate measurements discussed in [RFC7497], it would be beneficial for a Session-Reflector to respond with asymmetrical test packets: packets whose length is not symmetrical to the test packet sent by the Session-Sender and/or packets that are not sent in direct response to a packet received from a Session-Sender. The optional extension defined in this document gives operators the tools to create such asymmetrical packets between a Session-Sender and a Session-Reflector.

Measurement of performance metrics in a multicast network using an active measurement method (Section 3.4 of [RFC7799]) has specific challenges compared to what operators experience monitoring in a unicast network. This document analyzes these challenges and specifies procedures and STAMP extensions to achieve more efficient measurements with a lesser impact on a network.

2. Conventions Used in This Document

2.1. Terminology

This document uses terms defined in [RFC8762], specifically Session-Sender, Session-Reflector, and symmetrical packets.

This document uses terms defined in [RFC8972], specifically STAMP Session Identifier (SSID), STAMP TLV Flags, and Sub-TLVs.

This document uses terms defined in [RFC7497], specifically In-Service and Out-of-Service.

In this document, "asymmetrical packets" has two meanings, depending on the context. The first aspect is asymmetry in packet size between a packet sent by a Session-Reflector and the packet it received from the Session-Sender. The second aspect is asymmetry in the number of packets the Session-Reflector transmits in response to receiving a single STAMP-Test packet.

In this document, a multicast network means a communication network model where a sender transmits a single packet addressed to a multicast group, and the network delivers copies of that packet to multiple receivers that have joined the group.

2.2. Acronyms

- CE: Congestion Experienced
- ECN: Explicit Congestion Notification
- EUI: Extended Unique Identifier
- MAC: Media Access Control
- STAMP: Simple Two-way Active Measurement Protocol
- TLV: Type-Length-Value

2.3. Requirements Language

The key words "MUST", "MUST NOT", "REQUIRED", "SHALL", "SHALL NOT", "SHOULD", "SHOULD NOT", "RECOMMENDED", "NOT RECOMMENDED", "MAY", and "OPTIONAL" in this document are to be interpreted as described in BCP 14 [RFC2119] [RFC8174] when, and only when, they appear in all capitals, as shown here.

3. Reflected Test Packet Control TLV

This section defines an additional optional STAMP extension, the Reflected Test Packet Control TLV, and an additional bit flag in the STAMP TLV Flags field. The format of this TLV is presented in [Figure 1](#).

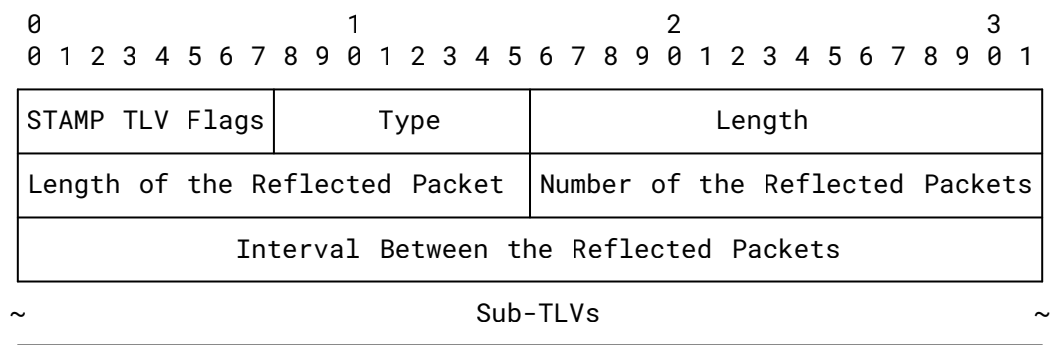


Figure 1: Reflected Test Packet Control TLV Format

The descriptions of the fields are as follows:

STAMP TLV Flags: A one-octet field [RFC8972].

Type: A one-octet field that identifies the Reflected Test Packet Control TLV. This field is set to 12 (Section 6.1).

Length: A two-octet field. The value is variable and **MUST NOT** be smaller than 12 octets.

Length of the Reflected Packet: A two-octet field. The value is an unsigned integer that is the requested length of a reflected test packet in octets.

Number of the Reflected Packets: A two-octet field. The value is an unsigned integer that is the number of reflected test packets that the Session-Reflector is requested to transmit in response to receiving a STAMP-Test packet with the Reflected Test Packet Control TLV.

Interval Between the Reflected Packets: A four-octet field. The value is an unsigned integer set to the interval in nanoseconds between the transmission of the consecutive reflected test packets in response to receiving a STAMP-Test packet with the Reflected Test Packet Control TLV.

Sub-TLVs: An optional field that includes additional information communicated by a Session-Sender.

Also, an additional STAMP TLV flag [RFC8972], the Conformant Reflected Packet, has been allocated by IANA in the "STAMP TLV Flags" registry (Section 6.2): the one-bit C flag (3). A Session-Sender **MUST** zero this flag on transmission, and the Session-Reflector **MUST** ignore its value on the receipt of a STAMP-Test packet with a STAMP TLV.

A Session-Sender **MAY** include the Reflected Test Packet Control TLV in a STAMP test packet. If the received STAMP-Test packet includes the Reflected Test Packet Control TLV, the Session-Reflector **MUST** transmit a sequence of reflected test packets according to the following rules:

- The length of the reflected test packet **MUST** be the largest of:
 - a. The length of a base Session-Reflector packet in the mode (unauthenticated or authenticated) of the received STAMP-Test packet, as defined in Section 4.3 of [RFC8762], including all STAMP extension TLVs [RFC8972] present in the received STAMP-Test packet but excluding any Extra Padding TLVs. The rationale to exclude any Extra Padding TLVs present in combination with the Reflected Test Packet Control TLV is to support a scenario in which a Session-Reflector is requested to transmit a sequence of packets shorter than the received STAMP packet.
 - b. The value in the Length of the Reflected Packet field of the Reflected Test Packet Control TLV aligned at a four-octet boundary.

In a case where the length of the reflected packet calculated by this rule is longer than the length of the reflected packet calculated by the rules in Section 4 of [RFC8972], the Session-Reflector **MUST** use the Extra Padding TLV (Section 4.1 of [RFC8972]) to increase the length of the reflected test packet. If the calculated length of the reflected packet exceeds the maximum transmission unit (MTU) of the interface to reach the Session-Sender, the Session-Reflector **MUST** set the

Conformant Reflected Packet STAMP TLV flag ([Section 6.2](#)) to 1 and **MUST** transmit a single reflected packet of the length equal to the MTU of the egress interface. Otherwise, the Session-Reflector **MUST** set the C flag to 0 in each reflected test packet.

The number of reflected test packets in the sequence **MUST** equal the value of the Number of the Reflected Packets field.

If the value of the Number of the Reflected Packets field is greater than 1, the interval between the transmission of two consecutive reflected packets in the sequence **MUST** be equal to the value in the Interval Between the Reflected Packets field in nanoseconds. To prevent excessive congestion caused by reflected packets, a Session-Reflector that supports the Reflected Test Packet Control TLV **MUST** enforce limits on both the data rate (bytes per second) and the total data volume (bytes) of the STAMP payload it generates in response to an incoming test packet. If a test packet is received that would generate traffic that exceeds either of these limits, the Session-Reflector **MUST** set the C flag ([Section 6.2](#)) to 1 and **MUST** transmit a single reflected packet of the length calculated by the rules listed above. Otherwise, the Session-Reflector **MUST** set the C flag to 0 in each reflected test packet.

If the Number of the Reflected Packets field is set to 0, the Session-Reflector **MUST NOT** send any reflected packets. Furthermore, in this case, the Session-Reflector **SHOULD** discard the received STAMP-Test packet. However, a local policy **MAY** override this default behavior and specify an alternative handling. Note that this behavior of the Session-Reflector is demonstrated when the Control Code Flags field of the Return Path Control Code sub-TLV ([Section 4.1.1](#) of [\[RFC9503\]](#)) is set to No Reply Requested. If this is the intended behavior, use of the Return Path TLV is preferable.

Each reflected test packet in the sequence is formed according to [Section 4.3](#) of [\[RFC8762\]](#).

As defined above, there are two cases when a Session-Reflector will set the C flag in the reflected packet. To disambiguate which case led to the C flag being set to 1, an implementation of a Session-Sender can determine the cause as follows:

- If the length of the received reflected STAMP packet is less than the value of the Length of the Reflected Packet field, the requested length exceeds the MTU of the egress interface of the Session-Reflector.
- If the length of the received reflected STAMP packet equals the value of the Length of the Reflected Packet field, the requested data rate and/or the data volume exceed the limits set at the Session-Reflector.

3.1. Address Group Sub-TLVs

A multicast network that uses an active performance measurement method for In-Service rate estimation **MUST** include a rate control mechanism that bounds and regulates the generation of measurement packets. Because multicast replication can amplify probe traffic across the distribution tree, uncontrolled probe emission risks introducing congestion, altering traffic asymmetry, or otherwise perturbing the conditions being measured. The rate control mechanism **MUST** ensure that probe traffic remains non-intrusive, predictable, and consistent with the

operational characteristics of the multicast topology. Aligning probe generation behavior with the timing and packet selection semantics of the asymmetric packet measurement method makes it possible for observations collected at receivers to remain valid and comparable. To allow for deployment on networks with different characteristics (i.e., latency, throughput, etc.), implementations **SHOULD** provide operators with the ability to configure rate limits and pacing parameters that prevent excessive or uneven probe replication while still enabling statistically meaningful measurement samples.

3.1.1. Layer 2 Address Group Sub-TLV

An optional Layer 2 Address Group sub-TLV is a variable-length sub-TLV that includes a Layer 2 Address Group Mask and Address Group fields used by the Session-Sender to select the Session-Reflectors for a response. The Layer 2 Address Group sub-TLV can convey EUI-48 (Extended Unique Identifier), EUI-64 [IEEE-802.3-2022], and a 16-bit short address for local identification within a Personal Area Network [IEEE-802.15.4-2024]. The format of the Layer 2 Address Group sub-TLV is presented in Figure 2.

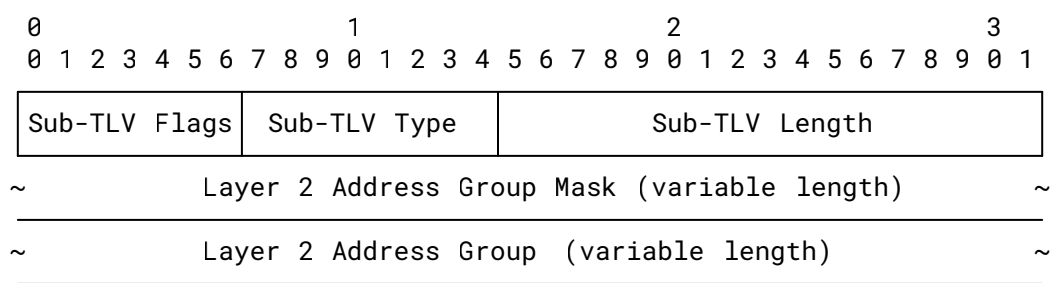


Figure 2: Layer 2 Address Group Sub-TLV Format

Where:

- Sub-TLV Flags: An eight-bit field. The format, values, and interpretation of flags are as defined for STAMP TLV Flags [RFC8972]. Flag values are taken from the "STAMP TLV Flags" registry [IANA-STAMP].
- Sub-TLV Type: A one-octet field. IANA has assigned value 10 (Section 6.3).
- Sub-TLV Length: A two-octet field whose value equals the length of the Value field of the Layer 2 Address Group sub-TLV in octets. Because the lengths of the Layer 2 Address Group Mask and Layer 2 Address Group fields **MUST** be equal, valid values for the Sub-TLV Length are 4, 12, and 16. Any other value **MUST** be considered by the Session-Reflector as a malformed sub-TLV.

The Value field of the Layer 2 Address Group sub-TLV consists of the following fields:

Layer 2 Address Group Mask: A field that represents the bitmask to be applied to all MAC addresses associated with the Session-Reflector. The length of the field is 1/2 the value of the sub-TLV Length field.

Layer 2 Address Group: A field that represents the group to which this TLV is addressed. The length of the field is 1/2 the value of the sub-TLV Length field.

If the Session-Reflector applies the value of the Layer 2 Address Group Mask field (using a bitwise AND) to any of its MAC addresses with the same length and the result is equal to the value of the Layer 2 Address Group field, then the Session-Reflector **MUST** stop processing the Layer 2 Address Group sub-TLV and continue processing the received test packet. If no matches are found, the Session-Reflector **MUST** stop processing the received packet.

3.1.2. Layer 3 Address Group Sub-TLV

An optional Layer 3 Address Group sub-TLV is a variable-length sub-TLV that includes the IP Prefix and IP Prefix Length fields used by the Session-Sender to select the Session-Reflectors for a response. The format of the Layer 3 Address Group sub-TLV is presented in [Figure 3](#).

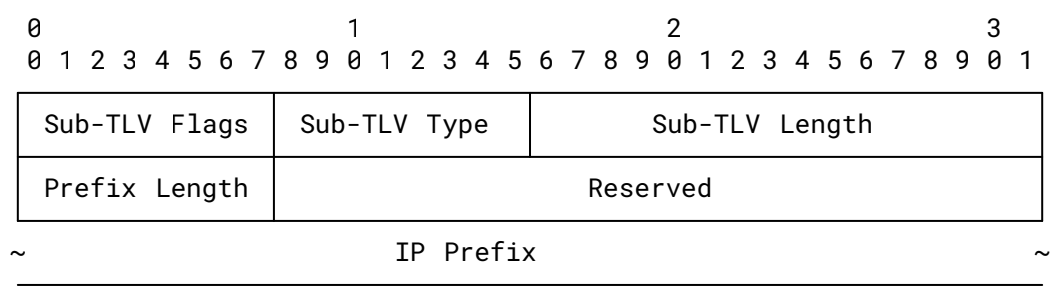


Figure 3: Layer 3 Address Group Sub-TLV Format

Where:

- Sub-TLV Flags:** An eight-bit field. The format, values, and interpretation of flags are as defined for STAMP TLV Flags [\[RFC8972\]](#). Flag values are taken from the "STAMP TLV Flags" registry [\[IANA-STAMP\]](#).
- Sub-TLV Type:** A one-octet field. IANA has assigned value 11 ([Section 6.3](#)).
- Sub-TLV Length:** A two-octet field whose value equals either 8 (if the IP Prefix is the prefix for an IPv4 address) or 20 (if the IP Prefix is the prefix for an IPv6 address). Any other value **MUST** be considered by the Session-Reflector as a malformed sub-TLV.
- The Value field of the Layer 3 Address Group sub-TLV consists of the following fields:

Prefix Length: A one-octet unsigned integer field that contains the length, in bits, of the prefix of the value in the IP Prefix field.

Reserved: A three-octet field. The field **MUST** be zeroed on transmission and ignored on receipt.

IP Prefix: A variable-length field. The length of the field is four octets if the IP Prefix is the prefix for an IPv4 address or 16 if the IP Prefix is the prefix for an IPv6 address.

When processing this sub-TLV, the Session-Reflector will construct an IP mask according to the value, *n*, in the Prefix Length field. The IP mask will be an IP address (of the family specified by the value of the sub-TLV Length field, according to the semantics above) where the *n* most-significant bits are set to 1 and all other bits are set to 0. Once the mask is constructed, if the Session-Reflector applies it (using a bitwise AND) to any of its IP addresses of the same family and the result is equal to the value in the IP Prefix field, then the Session-Reflector **MUST** stop processing the Layer 3 Address Group sub-TLV and continue processing the received test packet. If no matches are found, the Session-Reflector **MUST** stop processing the received packet.

4. Operational Considerations

4.1. Rate Measurement

[RFC7497] defines the problem of access rate measurement in access networks. One of the essential requirements identified for a test protocol is the ability to control packet characteristics on the tested path, such as asymmetric rate and asymmetric packet size. The Reflected Test Packet Control TLV, defined in Section 3, conforms to the requirements for measuring access rate by providing optional controls of the number of reflected test packets, the size of the reflected packet(s), and the time interval, i.e., rate, in transmitting the sequence of the reflected test packets. The access rate metric and method of access rate measurement are out of the scope of this document. The UDP Speed Test (see [RFC9097] and [RFC9946]) also allows for the measurement of access bandwidth.

4.1.1. Operational Considerations for Performing Rate Measurement

General considerations for using a testing protocol for rate measurement are documented in Section 7 of [RFC7497]. These considerations are specific for In-Service and Out-of-Service rate measurement. In the Out-of-Service testing, an operator may use a very high traffic rate and/or volume (i.e., high values for the Length of the Reflected Packet and/or Number of the Reflected Packets fields, and/or low values for the Interval Between the Reflected Packets field of the Reflected Test Packet Control TLV) to create congestion in the bottleneck. However, when performing In-Service rate testing, an operator may start with a low rate and/or volume and gradually increase them with each transmitted Reflected Test Packet Control TLV.

A service subscriber performing extensive rate measurements on the operational network **SHOULD** consider bullet item 6 in Section 11 of [RFC9946] and be mindful of limits placed on their service by the Service Provider. In particular, active measurement can lead to the generation of data volumes that may cause those performing the test to violate service-level agreements with their Service Provider.

4.2. Active Performance Measurement in a Multicast Environment

For performance measurements using STAMP in a multicast environment, a Session-Sender is expected to be the root and Session-Reflectors are the leaves of the same multicast distribution tree. The mechanism of constructing the multicast tree is outside the scope of this document.

According to [RFC8972], a STAMP Session is demultiplexed by a Session-Reflector by the tuple that consists of source and destination IP addresses, source and destination UDP port numbers, or the source IP address and STAMP Session Identifier. That is also the case when monitoring the performance of a multicast flow, despite the fact that the destination IP address is a multicast address. Therefore, there is no special behavior defined for a Session-Reflector upon receiving a STAMP-Test packet over a multicast tree. It processes the packet according to [RFC8762] and [RFC8972]. The Session-Reflector **MUST** use the source IP address of the received STAMP-Test packet as the destination IP address of the reflected test packet and **MUST** use one of the IP addresses associated with the node as the source IP address for that packet. As a result, a Session-Sender may receive multiple replies from multiple counterpart Session-Reflectors. Such a Session-Sender may include a Reflected Test Packet Control TLV and include either a Layer 2 Address Group sub-TLV or a Layer 3 Address Group sub-TLV to limit the Session-Reflectors that respond.

The multicast environment itself could be configured to help alleviate the possibility that network congestion may occur if a single test packet generates a large number of concurrent replies, all directed to the same endpoint. Depending on the multicast implementation, adding the Reflected Test Packet Control TLV could allow the multicast environment to limit the number of replies by modifying the Reflected Test Packet Control TLV sub-TLV values of any STAMP packets it sees, allowing replies only from reflectors that are:

- Randomly selected, by specifying a Layer 2 Address Group sub-TLV: for example, setting the EUI-48 Address Group Mask to 0xF and the EUI-48 Address Group to 0x1. As a result, only 1 out of 16 reflectors will reply;
- Hosted on a specific vendor Network Interface Card, by specifying a Layer 2 Address Group sub-TLV with the EUI-48 Address Group Mask set to 0xFFFFF000000; and
- Belonging to specific IP networks, for example, a subnet dedicated to IPv6-over-IPv4 encapsulation, by specifying the appropriate Layer 3 Address Group sub-TLV.

Multicast traffic is also intrinsically asymmetrical. The upstream (source-to-receiver) direction typically dominates, while the return path receives limited attention because multicast communication is primarily one-to-many and generates comparatively little downstream or receiver-to-source traffic. The value of the Length of the Reflected Packet field can be used to ensure that the reflected packet transports all the timestamps and requested information, which are crucial for the underlying measurement, but is as short as possible so as not to flood the network with useless data.

4.3. Using Reflected Test Packet Control TLV in Combination with Other TLVs

[RFC9503] defines the Return Path TLV that, when used in combination with the Return Address Sub-TLV, allows a Session-Sender to request the reflected packet be sent to a different address from the Session-Sender one. These STAMP extensions could be used in combination with the Reflected Test Packet Control TLV, defined in this document, to direct the reflected STAMP-Test packets to a collector of measurement data (according to [RFC7594]) for further processing and network analytics. An example of the use case is a multicast scenario when, for example, the Session-Sender is close to the actual multicast source (such as a camera transmitting live video) so that the test packets follow the same path as the video stream packets in one direction but the reflected test packets follow another to a destination where the data would be analyzed.

For compatibility with [RFC9503], a Session-Sender **MUST NOT** include a Return Path Control Code sub-TLV with the Control Code Flags set to No Reply Requested in a test packet that also contains a Reflected Test Packet Control TLV with a non-zero value. A Session-Reflector that supports both TLVs **MUST** set the U flag to 1 in both the Return Path and Reflected Test Packet Control TLVs within the reflected STAMP packet. Furthermore, the Session-Reflector **SHOULD** log a notification to inform an operator about the misconstructured STAMP packet.

The Reflected Test Packet Control TLV can be combined with the Class of Service TLV [RFC8972] to augment rate testing or testing in a multicast network that monitors the consistency of Differentiated Services Code Point and ECN values in forward and reverse directions of the particular STAMP-Test session.

5. Security Considerations

Security considerations discussed in [RFC7497], [RFC8762], [RFC8972], and [RFC9503] apply to this document. Furthermore, spoofed STAMP-Test packets with the Reflected Test Packet Control TLV can be exploited to conduct a Denial-of-Service (DoS) attack. Hence, implementations **MUST** use an identity protection mechanism. For example, the Session-Reflector may verify the information about the source of the STAMP packet against a pre-defined list of trusted nodes. Furthermore, an implementation that supports this specification **MUST** provide administrative control of support of the Reflected Test Packet Control TLV on a Session-Reflector with it being disabled by default. Also, either the STAMP authentication mode [RFC8762] or the HMAC (Hashed Message Authentication Code) TLV [RFC8972] **SHOULD** be used for a STAMP-Test session containing the Reflected Test Packet Control TLV. Note that if integrity protection is enabled, any in-path modification will cause verification to fail unless the modifying element is within the trust boundary and can recompute the integrity check.

Furthermore, a DoS attack using the Reflected Test Packet Control TLV might target the STAMP Session-Reflector by overloading it with test packet reflection, e.g., minuscule intervals and/or an excessive number of concurrent test sessions. To mitigate that, a Session-Reflector implementation that supports the new TLV **MUST** provide a mechanism to limit the reflection rate and volume of STAMP-Test packets (see Section 3 for a detailed discussion).

Considering the potential number of reflected packets generated by a single test packet sent to a multicast address, parameters in the first STAMP-Test packet with the Reflected Test Packet Control TLV **MUST** be selected conservatively. Consider the Number of the Reflected Packets field value set to one. As a result, a Session-Sender, by counting the packets reflected after originating a first STAMP-Test packet with the Reflected Test Packet Control TLV, can evaluate the load caused by using the Reflected Test Packet Control TLV in which more than a single reflected packet to the same multicast destination is requested. To further mitigate the risk of using the Reflected Test Packet Control TLV in a multicast network, a Session-Sender **SHOULD** sign packets using the HMAC TLV when sending such messages in unauthenticated mode [RFC8762]. But even with the HMAC TLV, the Reflected Test Packet Control TLV could be exploited by a replay attack. To mitigate that risk, a STAMP Session-Reflector **SHOULD** use the value of the Sequence Number field [RFC8762] of the received STAMP-Test packet. If that value compared to the received value in the previous test packet of the same STAMP-Test session is not monotonically increasing, then the Session-Reflector **MUST** respond with a single reflected packet, setting the U flag to 1 [RFC8972]. That may not indicate a replay attack, but there is packet re-ordering or packet duplication in the network. An operator can use other diagnostic methods to characterize and localize the problem. An implementation of the Session-Reflector can use the Serial Number Arithmetic [RFC1982] or any of the other methods to verify the correct ordering of test packets.

A Session-Sender **SHOULD NOT** send the next STAMP-Test packet with the Reflected Test Packet Control TLV before the Session-Reflector is expected to complete the transmission of all reflected packets in response to the Reflected Test Packet Control TLV in the previous test packet. In some scenarios, the Reflected Test Packet Control TLV might induce congestion on the transient bottleneck. Section 10 of [RFC9097] specifies security requirements for capacity measurements with asymmetric UDP loads.

When planning In-Service capacity measurement, operators **SHOULD** follow recommendations formulated in Sections 3 and 7 of [RFC7497]. If the underlay network is ECN-capable, a Session-Reflector may receive STAMP-Test packets with the ECN field marked as Congestion Experienced (CE). ECN markings provide an indication of incipient congestion rather than packet loss. However, the interpretation of what constitutes "significant congestion" and the operational thresholds for reacting to ECN-CE depend on the specific deployment, service objectives, and operator policy. Operators should be aware that In-Service capacity measurements may influence congestion conditions, potentially contributing to ECN-CE marking in the network. Implementations and operational procedures **SHOULD** ensure that the use of STAMP for In-Service measurement does not unintentionally degrade data traffic or lead to misinterpretation of ECN-related congestion signals. Appropriate thresholds and mitigation actions remain deployment-specific and **SHOULD** be guided by operator policy and network performance objectives.

Furthermore, Section 3.1.5 of [RFC8085] determines that a UDP congestion control **SHOULD** respond quickly to experienced congestion and account for loss rate and response time when choosing a new rate. And Section 8.1 of [RFC9097] specifies the load rate adjustment algorithm with its sample pseudocode offered in Appendix A of [RFC9097].

6. IANA Considerations

6.1. Reflected Test Packet Control TLV Type

IANA has assigned a new value for the Reflected Test Packet Control TLV in the "STAMP TLV Types" registry under the "Simple Two-way Active Measurement Protocol (STAMP) TLV Types" registry group as follows:

Value	Description	Reference
12	Reflected Test Packet Control	RFC 10052

Table 1: Reflected Test Packet Control TLV Type

6.2. Conformant Reflected Packet STAMP TLV Flag

IANA has allocated a bit position for the Conformant Reflected Packet STAMP TLV flag in the "STAMP TLV Flags" registry under the "Simple Two-way Active Measurement Protocol (STAMP) TLV Types" registry group as follows:

Bit position	Symbol	Description	Reference
3	C	Conformant	RFC 10052

Table 2: Conformant Reflected Packet STAMP TLV Flag

6.3. Layer 2 and Layer 3 Address Group Sub-TLV Types

IANA has assigned values for the Layer 2 Address Group and Layer 3 Address Group sub-TLV Types in the "STAMP Sub-TLV Types" registry under the "Simple Two-way Active Measurement Protocol (STAMP) TLV Types" registry group as follows:

Value	Description	TLV Used	Reference
10	Layer 2 Address Group	Reflected Test Packet Control	RFC 10052
11	Layer 3 Address Group	Reflected Test Packet Control	RFC 10052

Table 3: STAMP Sub-TLV Types for the Reflected Test Packet Control TLV

7. References

7.1. Normative References

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Authors' Addresses

Greg Mirsky

Ciena Corporation

Email: gregimirsky@gmail.com

Ernesto Ruffini

OutSys

Email: eruffini@outsys.org

Henrik Nydell

Cisco Systems

Email: hnydell@cisco.com

Richard Foote

Nokia

Email: footer.foote@nokia.com

Will Hawkins

University of Cincinnati

Email: hawkinsw@obs.cr